

Two-sample reading budget: same expert and context packing

CFPerm prototype

Not TMLE, not causal, not online learning. W is a batch tag; $\hat{e} \approx P(W=1 \mid X)$. Adaptation: freeze π from a domain pair, then feed a *single* reader the concatenated *raw* blocks $E = E(\pi)$. The reader is not updated from the query stream; π is not updated either. An opinion pool of $\hat{e}_m$ is a mixture of scores, not this map.

1 Raw block + one reader

Let $E = E(\pi)$. The observation is $\text{pack}(x) = \text{concat}_{m \in E} \text{serialize}(x_{B_m})$. Output is $f(\text{pack}(x))$ with one f . Dropped blocks are absent. This restricts the σ -algebra of a frozen reader. It is not Hedge, not a bandit over specialists, and not a per-query update of π .

A query updates E only: $r = \lambda\pi + (1 - \lambda)s(x)$, then $E_{\text{query}} = \text{pack}(r)$. Stage A evaluates block prediction $\hat{e}_m$; Stage B evaluates the adapted read after that update. $\lambda = 1$ is prior-only; $\lambda = 0$ is a hard query gate (often noisier).

2 Same expert versus packing

A *same expert* is $\hat{m} = \arg \max \pi$, then $f \in \mathcal{G}_{\hat{m}}$ (functions of $X_{B_{\hat{m}}}$ only) for every query. That template does not depend on x . An instance gate $\hat{m}(x) = \arg \max s_m(x)$ is a different object.

Context packing is one reader on concatenated raw blocks $E = E(\pi)$, not an opinion pool of $\hat{e}_m$. Disabled channels are omitted, not emitted as empty headers. Abstain means the gap is not localized: do not drop a block (pack all, or HITL). Forced top-1 on a diffuse shift discards signal.

3 Specialist $\hat{e}_m$ (one block, all rows)

4 Packed reader (one RF on concat E)

Inject valence with a wide text block: VIMP hires the wide specialist; π hires valence and the packed reader matches the GT block. Diffuse equal shift: pack-all beats one specialist. Shuffle valence: hit collapses.

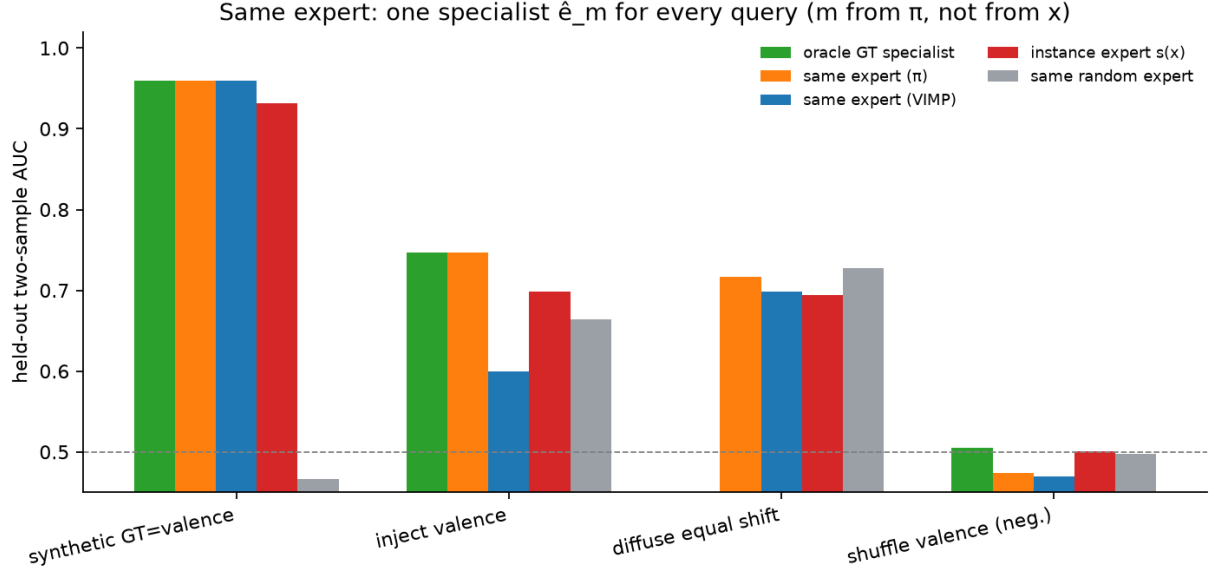


Figure 1: Frozen specialist AUC. π picks one m for the whole test set.

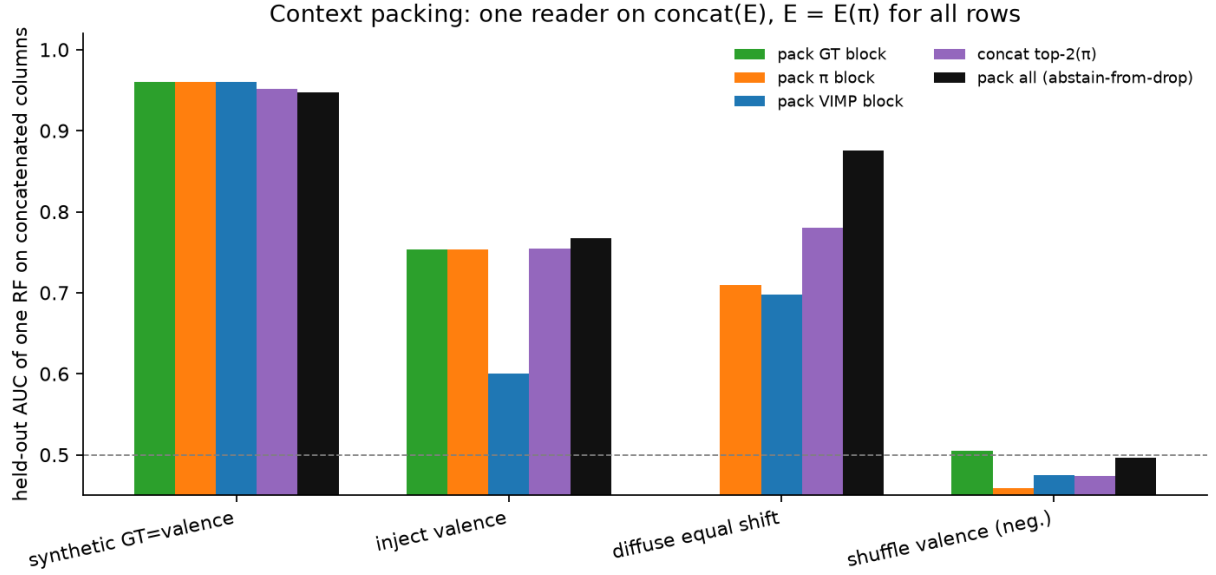


Figure 2: Column-concat of $E(\pi)$. Pack-all is the abstain-from-dropping action.

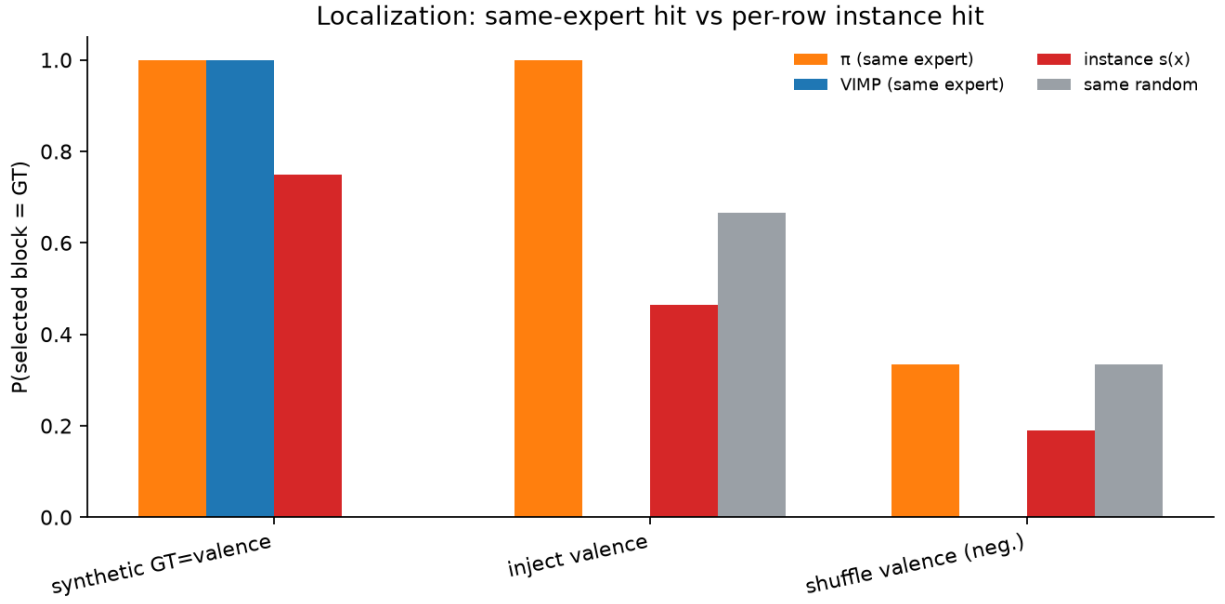


Figure 3: Localization hit. Same-expert hit is $1\{\hat{m} = m^*\}$, not a per-row mix.

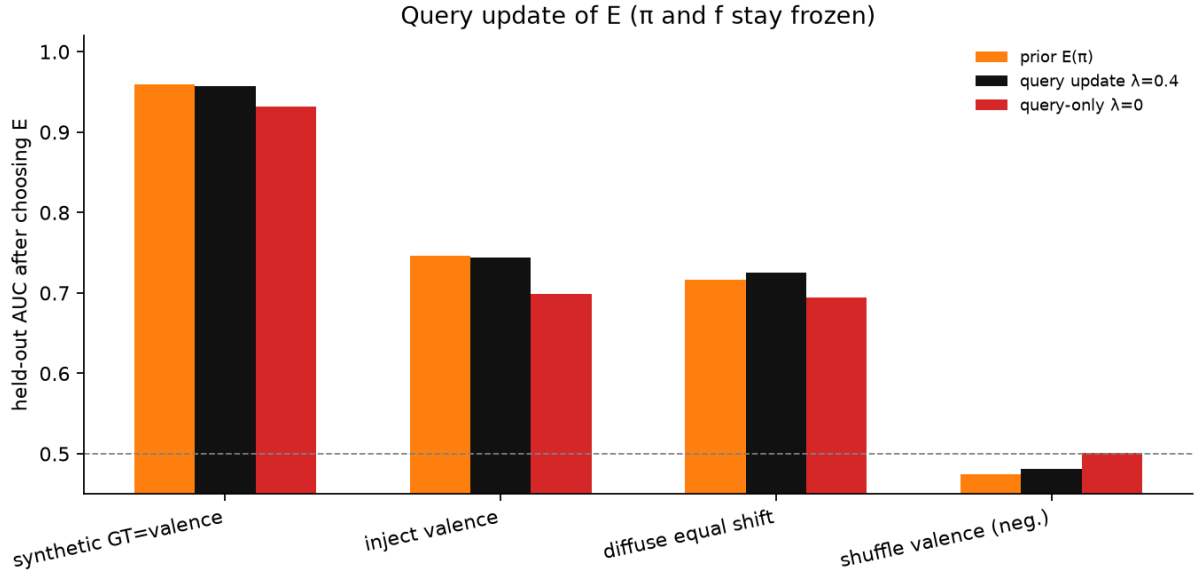


Figure 4: Query update of E : prior vs shrinkage vs s -only. π and f stay frozen.